

Quite Universal Circuit Simulator: A Free Mixed Signal Electronic Circuit Simulator

Being a free electronic circuit simulator, this is an efficient tool for an electronics engineer to simulate analogue and digital systems as well as analyse their system behaviour. Try it out in this month's EFY Plus DVD

JAI SACHITH PAUL

Circuit simulation is a critical step in electronic product design. By simulation a design, an engineer will know the behaviour of the circuit on various test cases even before practically implementing it. Getting the design right at the first implementation saves a lot of time and cost for the manufacturer. This, in turn, helps the manufacturer to reduce the time to market.

Quite universal circuit simulator (Qucs) is a free open source integrated circuit simulator with a smart graphical user interface (GUI). The GUI in Qucs is based on Qt, which is a cross-platform application framework. We can run AC, DC and transient analysis, in addition to VHDL/Verilog simulations, very easily with the help of this circuit simulator. Apart from this, we can easily import any existing SPICE model for our circuit simulation.

Infobytes

- Qucs is currently developed for GNU/Linux systems using GNU autotools.
- Also available natively on X86 Windows (XP/Vista/7/8) systems and experimentally on X86-64 versions of these. It also gets successfully compiled on other operating systems like Solaris and NetBSD.
- Latest stable release: 0.0.18 on August 31, 2014.

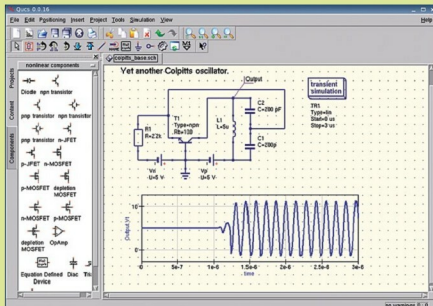


Fig. 1: Schematic and data visualisation

Functions of Various Tool Suites

Tool suite	Function
GUI	Creates schematics and VHDL code, sets up simulations and provides simulation results.
Text editor	Displays netlists and simulation logging information.
Filter synthesis application	Helps in designing various types of filters.
Backend analogue simulator	Simulates the previously set-up schematic. It takes a netlist, checks it for errors, performs required simulation actions and, finally, produces a dataset.
Transmission line calculator	Used to design and analyse different types of transmission lines like microstrips, coaxial cables, etc.
Component library	Holds models of real-life devices such as resistors, diodes, opamps, etc.
Attenuator synthesis application	Designs various types of passive attenuators.
Command line-conversion program	Imports and exports datasets, netlists and schematics from and to other CAD/EDA software.